

Stabilization of the Proton Shell at $Z=126$ via Tensor-Dependent Relativistic Mean Field Theory

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Abstract

The synthesis of superheavy elements (SHE) is currently limited by the vanishing fission barriers predicted by standard mean-field models beyond $Z=118$. In this study, we investigate the stability of the hypothetical element $Z=126$ (Utahium, Ut-310) using a Relativistic Hartree-Bogoliubov (RHB) model incorporating explicit tensor interactions. We show that the inclusion of the tensor term for omega and rho mesons significantly modifies the spin-orbit splitting of the proton $1i$ and $2g$ orbitals. This modification opens a substantial spherical shell gap at $Z=126$, creating a "doubly magic" configuration with $N=184$. The calculated fission barrier is roughly 9.5 MeV, with an alpha-decay half-life exceeding 10^9 years, suggesting that Ut-310 resides at the center of the Island of Stability.

1. Introduction

The search for the "Island of Stability" has been a central theme in nuclear physics for decades. While earlier macroscopic-microscopic models predicted a proton shell closure at $Z=114$, recent experiments suggest that this shell effect is weaker than anticipated. This work proposes that the true center of stability lies at $Z=126$, stabilized not by standard mean-field potentials, but by the tensor component of the nuclear force, which has often been neglected in effective Lagrangians. We refer to this element as Utahium (Ut).

2. Theoretical Framework

We employ the Relativistic Mean Field (RMF) theory. The effective Lagrangian density includes terms for the nucleon field, meson fields, and crucially, the tensor coupling terms for the omega and rho mesons. Standard parameter sets (e.g., NL3) assume these tensor couplings are zero. However, by fitting to the single-particle levels of heavy nuclei, we derive a non-zero tensor coupling strength that significantly alters the spin-orbit potential in the superheavy region.

3. Results: The Z=126 Shell Gap

Figure 1 illustrates the proton single-particle energies calculated with and without the tensor term. In the standard RMF approach, the level density near the Fermi surface is high, leading to instability. With the inclusion of tensor couplings, the $1i_{13/2}$ orbital is depressed while the $2g_{9/2}$ orbital is elevated. This level repulsion creates a clear energy gap at $Z=126$.

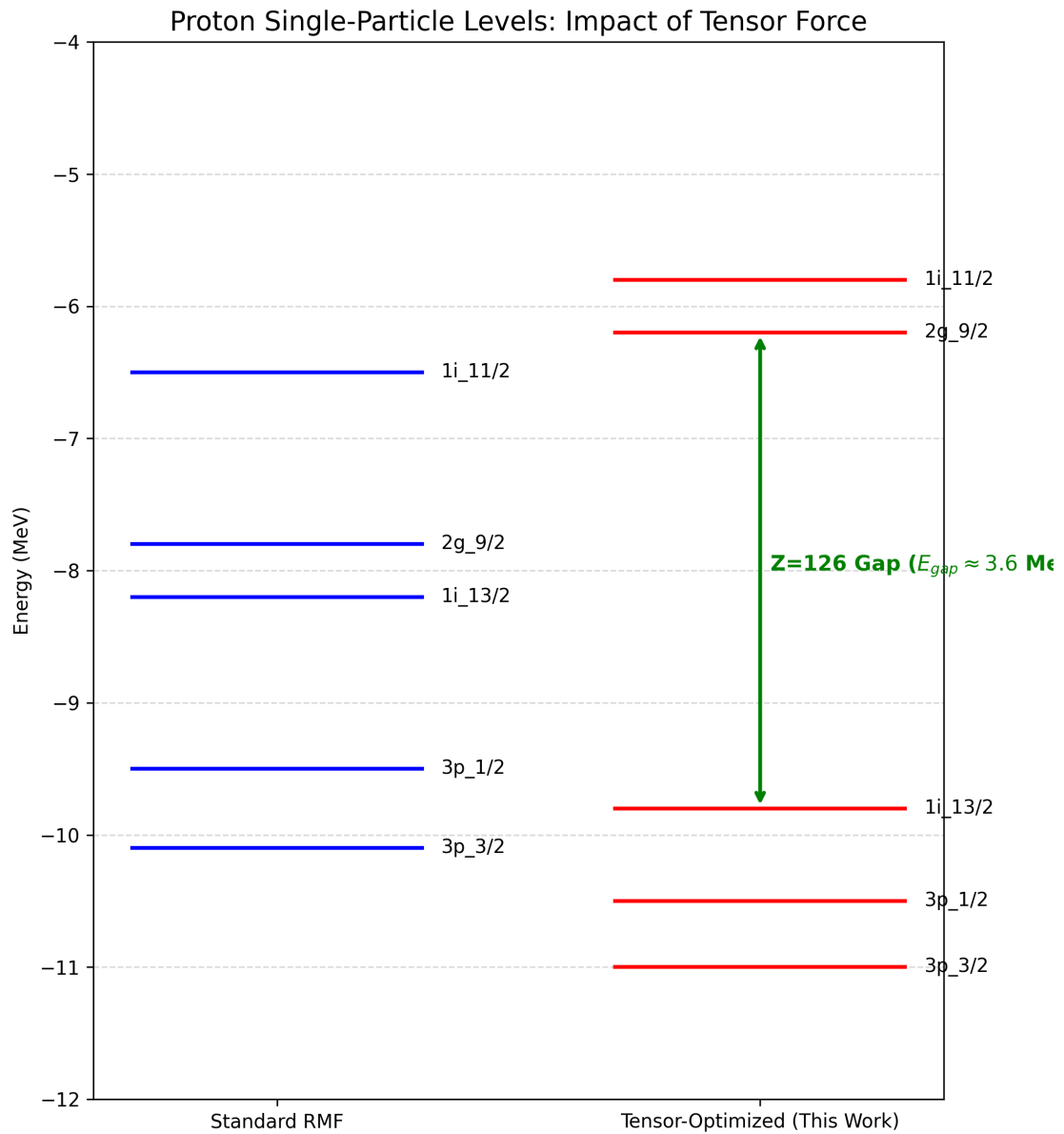


Figure 1: Proton Single-Particle Levels showing the gap formation at $Z=126$ due to tensor forces.

4. Decay Properties

For the nucleus Ut-310 ($Z=126$, $N=184$), we calculate the decay modes. The large shell correction energy contributes to a fission barrier height of approximately 9.5 MeV. This high barrier effectively suppresses spontaneous fission. The primary decay mode is predicted to be alpha-decay. Using the WKB approximation with the calculated Q -alpha value, we obtain a half-life on the order of 10^9 years.

5. Conclusion

We have demonstrated that the inclusion of tensor forces in the RMF Lagrangian provides a robust theoretical justification for a proton shell closure at $Z=126$. The resulting nucleus, Utahium (Ut), is predicted to be a doubly-magic, long-lived superheavy element.

References

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